Multidimensional Transport Poverty Index

A replicable multi-dimensional and multi-modal measure of transport poverty

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Abstract

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Keywords: keyword1, keyword2

1. Introduction

Transportation is an essential service which can condition access to almost all human activities, be them work, education, health or social life. When this access is limited by insufficient supply, high costs, or elevated exposure to systemic risks and externalities, the term Transport Poverty is used (Lucas et al., 2016). This phenomenon has direct

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consequences in social exclusion and territorial cohesion (Mejia-Dorantes and Murauskaite-Bull, 2022).

In the European context, this problem has increased its political relevance over the last decade. The European Pillar of Social Rights, in its 20th Principle, enshrines the right to access essential services, including transportation. Regulation (EU) 2023/955 of 10 May 2023, which established the Social Climate Fund (SCF), enshrined the first legally bounding definition of transport poverty at an European scale, defining it as "individuals' and households' inability or difficulty to meet the costs of private or public transport, or their lack of or limited access to transport needed for their access to essential socioeconomic services and activities, taking into account the national and spatial context" (Parliament and the Council of the European Union, 2023) .

Despite the clear empirical relevance of this phenomenon, there is difficulty in quantifying it in a systematic way. Specifically in Portugal, previous research has focused on punctual case studies (Pritchard et al., 2014) or on costly mobility surveys (INE, 2018), which can hardly be renewed at the cadence required by public policy cycles.

This project was thus developed in an attempt to bridge the existing methodological gap with the creation of a composite and intermodal tool - the Multidimensional Transport Poverty Index (IMPT) - that is replicable and scalable, as well as wholly based in public data, intended to assist and support public policy instruments, such as the European Union's Social Climate Fund (SCF) or Sustainable Urban Mobility Plans (SUMPs). The Lisbon Metropolitan Area (LMA) was employed as a case study for the tool's validation.

2. Literature review

2.1. Definitions of Transport Poverty

Transport systems are a fundamental part of modern society, and their absence is related to social exclusion - a state where an individual or a societal group is left out of participating in regular social activities in a given society (Pritchard et al., 2014) -, socio-territorial inequality and transport inequality - the differences in transport conditions for different groups of people (Hidayati et al., 2021). Despite the relation between poverty and transport conditions, these are also related to a wide range of social issues, with motorized vehicle users most contributing to externalities, but with vulnerable road users being disproportionately affected by them in health, space allocation and time consumption (Gössling, 2016). Moreover, lower-income households tend to not have financial conditions to have motorized transport or to reside in central areas, relegating them to live in areas distant from the city center, often poorly served by public transportation. (Fedorowicz et al., 2020) (Lopes et al., 2023).

These differences in transport conditions can lead to transport poverty, a concept which lacks a single universally accepted definition in scientific literature. This expression has been used to broadly describe inequities related with transportation access, but also in a more restricted way to refer to the economic inability to support transportation costs (Mattioli et al., 2018). However, as Lucas et al. (2016) states, different terminologies relating to transport poverty - terms such as “mobility poverty”, “accessibility poverty” or “transport disadvantage” - are used with “often very different, although also overlapping definitions”. Due to this lack of a universal definition and the multidimensional nature of transport poverty, its measurement and quantification poses a challenge (Levinson and King, 2020).

This way, the necessity to comprehend this concept and quantify it becomes more urgent. It is essential to define reliable indicators of transport vulnerability to map “public transport deserts” (Jiao and Dillivan, 2013) and low-accessibility areas in order to inform public intervention and to monitor the implementation of measures to improve these situations. A reliable and replicable index to measure transport poverty is thus fundamental to support public policy instruments.

Lucas et al. (2016) proposes the most comprehensive conceptual systematization of transport poverty, clearly defining previously different but overlapping terminology. They define transport poverty as a combination of the following four dimensions:

- Accessibility - the capacity to reach usual destinations and activities (work, education, health and so on).
- Mobility - the systemic lack of transportation that impacts trips.

- Affordability - the ability to meet the cost of transportation.
- Exposure to externalities (Safety) - the negative exposure to the transport system itself, such as disproportional rates of pollution or road crashes.

Recently, and based on the aforementioned concepts, Mejia-Dorantes and Murauskaite-Bull (2022) have proposed a definition that has been widely adopted: transport poverty is the inability of an individual or household to access essential services or opportunities due to transportation services that are inaccessible, inadequate, unaffordable, time-consuming or unsafe. This definition has been adapted and applied at the European scale, through Parliament and the Council of the European Union (2023) and a recent European Commission Report (European Commission. Directorate General for Employment and Inclusion, 2024). The key dimensions in the EC Report are synthesized as *Availability* (corresponding to Lucas' Mobility), *Accessibility* and *Affordability*, with exposure to externalities being included in the overarching concept of *Adequacy*.

In summary, the concept of transport poverty has evolved from a one-dimensional look at accessibility or affordability, to a composite approach that attempts to capture the multidimensional nature of this phenomenon.

During the last decade various interactive spatial tools based on indicators of accessibility and inequality have been created and made publicly available. Two of these are here highlighted, due to their conceptual or methodological influence and their proximity to the IMPT's objective.

2.2. Dimensions of Transport Poverty

The dimensions proposed by Lucas et al. (2016) continue to be the reference for composite indexes of transport poverty (Alonso-Epelde et al., 2023) (Mejia-Dorantes and Murauskaite-Bull, 2022). IMPT adopts these dimensions, implementing them through secondary data available in Portugal. Each of them is further explained thoroughly.

2.2.1. Accessibility

Accessibility refers to the ability to reach regular destinations such as employment, education, health and other essential services (Allen and Farber, 2019) (Unit, 2003).

2.2.2. Mobility

Mobility refers to the systemic deficits of the transport system, be them lack of infrastructure, lack of service frequency or long trip duration (Levinson and King, 2020) (Lucas et al., 2016) (Mejia-Dorantes and Murauskaite-Bull, 2022). This dimension captures different aspects of transport supply that accessibility is not able to, their core difference being that, while accessibility asks which places are reachable, mobility focuses on how easy it is to reach them.

2.2.3. Affordability

2.2.4. Safety

This dimension is commonly not included in transport poverty indexes (Kelly et al., 2023), being considered only in the most wide-ranging definitions of transport poverty (Lucas et al., 2016) (UN-Habitat). It refers to the exposure to externalities of the transport system, such as road crashes and pollution, which have disproportionate effects on the territory. For the IMPT, this dimension's inputs only included those related to road safety indicators, due to the inability to get open and granular data regarding environmental externalities.

A fifth dimension, digital literacy, was initially considered as a proxy for access to information and mobility services (Durand et al., 2022) (Orozco-Fontalvo et al., 2025) (Velaga et al., 2012). However, it was excluded from the final implementation of the index due to unavailability of data compatible with the scale of the lowest administrative division.

2.3. IMPT's Contribution

IMPT emerges in an attempt to create an index synthesizing these dimensions and providing a reliable measurement of transport poverty, bridging three gaps in the literature and in national public policy.

1. There is, in Portugal, the absence of an index that integrates the four aforementioned dimensions across multiple transport modes and is replicable at the lowest administrative level.

2. Existing indexes excessively depend on costly primary data that is difficult to monitor continuously. IMPT relies exclusively on secondary and open data, easily allowing for updates and monitoring over time.
3. There is a lack of operational tools that allow to translate academic literature on transport poverty into decision-support instruments for local governments, transport authorities, and generally public policy makers.

Furthermore, the IMPT incorporates four methodological variants, corresponding to different analytical positions:

- Contrasting IMPT: based on entropy, highlights the variables with the highest variation, better capturing territorial contrasts.
- Critical IMPT: based on the geometric mean, it non-linearly penalizes the dimensions with the worst performance in each area.
- Balanced IMPT: based on the arithmetic mean, allows for compensation between dimensions.
- Political IMPT: allowing for a personalized AHP weighting, it can be tailored to the decision-maker's preferences.

If `cite-method` is set to `citeproc` in `elsevier_article()`, then pandoc is used for citations instead of `natbib`. In this case, the `csl` option is used to format the references. By default, this template will provide an appropriate style, but alternative `csl` files are available from https://www.zotero.org/styles?q=elsevier. These can be downloaded and stored locally, or the url can be used as in the example header.

Here are two sample references: Pereira et al. (2021) INE (2018). If citing the other way (Lovelace et al., 2022).

3. Methodology

3.1. Case Study

The Lisbon Metropolitan Area (LMA) was applied as the case study for the IMPT. The LMA is comprised of two NUTS II regions (Grande Lisboa and Península de Setúbal), which encompasses 18 Municipalities and approximately 2,8 million inhabitants. This region presents territorial characteristics which make it particularly suitable for this analysis: a polycentric structure with a strong employment center in Lisbon, significant discrepancies between the public transport offer in the urban core and in the periphery, and a high modal share of private vehicles in a large part of the metropolitan area.

The analysis is focused on the lowest administrative division, the *freguesia* (parish), which is the most disaggregated geographical level that guarantees data coverage for all utilized sources. For the indicators that had a higher level of granularity, an analysis by grid level was also assembled. The LMA includes 124 parishes, which constitute the observation units of the IMPT.

3.1.1. Data Sources

Lastly, the following R packages were utilized in the data processing and creation of both the index and its interactive dashboard: `readr`, `readxl`, `openxlsx`, `lubridate`, `dplyr`, `tidyr`, `stringr`, `tidyverse`, `sf`, `terra`, `mapview`, `stplanr`, `h3jsr`, `r5r`, `accessibility`, `odjitter`, `tidytransit`, `GTFShift`, `FactoMineR`, `assertthat`, `piggyback`.

3.2. Each Dimension's Indicators

The IMPT is not only multidimensional, but also intermodal. In Table 1, the dimensions used to compute the index for each mode are specified.

```
modes_per_dimension <- data.frame(
  Dimension = c("Accessibility",
                "Mobility",
                "Affordability",
                "Safety"
                ),
  Mode = c("Driving, Public Transport, Walking, Cycling",
           "Driving, Public Transport, Walking, Cycling",
           "Driving, Public Transport",
           "Driving, Walking, Cycling"
```

```
          )
)
knitr::kable(modes_per_dimension)
```

```
Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
Use 'xfun::attr2()' instead.
See help("Deprecated")

Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
Use 'xfun::attr2()' instead.
See help("Deprecated")
```

Table 1: Modes evaluated in each dimension

Dimension	Mode
Accessibility	Driving, Public Transport, Walking, Cycling
Mobility	Driving, Public Transport, Walking, Cycling
Affordability	Driving, Public Transport
Safety	Driving, Walking, Cycling

3.2.1. Accessibility

The Accessibility dimension quantifies the capacity to reach key destinations within a given travel time threshold. The key destination categories, as well as their data sources, are summarized in table (…)

This dimension incorporates two metrics:

1. Cumulative number of opportunities: number of points of interest of a given category reachable within a certain amount of time.
2. Travel time to reach nearest opportunities: time required to reach the first n opportunities of a given category.

Indicators for this dimension were computed using the H3 grid (Uber Technologies, 2018) of resolution 8 (Pereira and Herszenhut, 2022), which corresponds to 3686 hexagons in the

LMA, with an edge length of 531 meters and an average hexagon area of approximately 737,327m^2.

3.2.1.1. Jittering.

Due to a lack of open access to fine data regarding job location, the most granular data obtained was an OD matrix at the parish level, provided by INE (2018), with centroid to centroid representations, "a common approach involving the simplifying assumption that all trip destinations and origins can be represented by (sometimes population weighted or aggregated) zone centroids" (Lovelace et al., 2022). However, in order to accurately compute accessibility and mobility measures, there was the need to have more disaggregated data regarding job locations, beyond the parish unit.

This way, jittering (Lovelace et al., 2022) was used as a proxy for this purpose, using building volumes as trip attractors, and this way weighting origins and destinations. This allowed for the disaggregation of trips between parishes to ODs with up to 50 trips.

3.2.1.2. Travel time matrix.

A travel time matrix was used as the basis for all accessibility measures. It was computed between each grid cell, over a network (Pereira et al., 2021) that was built considering:

- OpenStreetMap road network, exported from HOT (2026).
- General Transit Feed Specification (GTFS) for all operators in the LMA (valid at 04/02/2026).
- Terrain elevation profile (EU, 2026).

The matrix was computed using the function `r5r::travel_time_matrix()` with the following parameters: ()

3.2.1.3. Cumulative number of opportunities.

The cumulative number of opportunities was calculated using the function `accessibility::cumulative_cutoff()` considering the travel time matrices for each grid cell combination, the number of opportunities per grid cell, and the travel time cut-offs.

This measure was computed for all combinations of POIs and modes for time cut-offs of 5, 10, 15, 30, 45, 60, 75 and 90 minutes. However, as can be seen in Table 2, only selected combinations of modes/points of interest were included in IMPT's operationalization.

```r
accessibility_n_opportunities <- data.frame(
  "Point of Interest/Mode" = c("Healthcare Facilities",
                              "Basic Education",
                              "Green Spaces",
                              "Recreation",
                              "Groceries",
                              "Jobs"
                              ),
  "Driving or Public Transport" = c("30","30","5-10","5-10","15","30-45"),
  "Walking or Cycling" = c("15","15","15-30","15-30","15","30-45"),
  check.names = FALSE
)
knitr::kable(accessibility_n_opportunities)
```

```
Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
Use 'xfun::attr2()' instead.
See help("Deprecated")
```

```
Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
Use 'xfun::attr2()' instead.
See help("Deprecated")
```

Table 2: Travel time threshold (minutes) to compute cumulative number of opportunities

Point of Interest/Mode	Driving or Public Transport	Walking or Cycling
Healthcare Facilities	30	15
Basic Education	30	15

Table 2: Travel time threshold (minutes) to compute cumulative number of opportunities

Point of Interest/Mode	Driving or Public Transport	Walking or Cycling
Green Spaces	5-10	15-30
Recreation	5-10	15-30
Groceries	15	15
Jobs	30-45	30-45

The value computed represents the number of opportunities reachable within the defined travel time threshold. Values at parish and municipality level are aggregated considering a weighted mean of the number of opportunities reachable in each grid cell within that geographical unit, weighted by the population of those cells.

3.2.1.4. Travel time to reach closest opportunities.

This measure was computed using the function `accessibility::cost_to_closest()` , considering the travel time matrices for each grid cell combination and the number of opportunities in each grid cell. Its output corresponds to the travel cost, in minutes, to reach n opportunities of a certain category. Although calculations were performed for 1, 2 and 3 opportunities, only some of these were included in IMPT's operationalization, depending on the POI category:

- Healthcare, basic education and green spaces: 1 opportunity.
- Recreation, groceries and jobs: 3 opportunities.

3.2.2. Mobility

Mobility evaluates the quality of the transportation system as a service, independently of the accessibility of the destinations. The methodology to calculate its principal indicators is explained and detailed below.

3.2.2.1. Average commuting time.

Using the jittered (Section 3.2.1.1) OD pairs from INE (2018) (for commuting) and the travel time matrices, average commuting travel times are computed for each geographical unit and mode, weighted by number of trips. Trips with no feasible route within the constraints are assigned the maximum travel time (120 min) and counted separately as “not accomplishable” trips (PNA - Percentagem Não Alcançáveis).

3.2.2.2. Average number of commuting transfers.

For each commuting OD pair, the minimum number of transfers required for each public transport route (for 0, 1, 2, and 3 transfers) is assigned to each OD pair, and population-weighted mean transfers are computed per geographical unit. The result is the average minimum amount of transfers to commute to work using public transport.

3.2.2.3. Public transport headways and waiting times.

Stop-level service frequencies were computed using `tidytransit::get_stop_frequency()` (Poletti et al., 2026) for four different time windows, as can be seen in

```r
headway_periods <- data.frame(
  Period = c("Weekday peak",
             "Weekday full day",
             "Weekday night",
             "Weekend"
             ),
  Day = c("Wednesday 04/02/2026","Wednesday 04/02/2026","Wednesday 04/02/2026","Sunday 08/02/2
  "Time window" = c("08:00-09:00","00:00-48:00","22:00-23:00","10:00-11:00"),
  check.names = FALSE
)
knitr::kable(headway_periods)
```

```
Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
Use 'xfun::attr2()' instead.
See help("Deprecated")
```

```
Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
```

```
Use 'xfun::attr2()' instead.
See help("Deprecated")
```

Table 3: Public transport service frequency periods observed

Period	Day	Time window
Weekday peak	Wednesday 04/02/2026	08:00-09:00
Weekday full day	Wednesday 04/02/2026	00:00-48:00
Weekday night	Wednesday 04/02/2026	22:00-23:00
Weekend	Sunday 08/02/2026	10:00-11:00

A 500-meter buffer per stop is used to allocate stops to BGRI (INE, 2021) population centroids. Results are aggregated to geographical units as population-weighted mean headway and waiting time. Additionally, frequency reduction factors were computed to quantify the reduction in service during night and weekend periods.

3.2.2.4. Public transport population coverage.

3.2.2.5. Ratio of pedestrian and cycling infrastructure.

The pedestrian infrastructure ratio is the proportion of road length that includes pedestrian infrastructure, computed using OpenStreetMap data:

$$R_{ped} = \frac{pedestrian\ path\ length}{road\ length}$$

Road classification uses the highway tags: `motorway`, `trunk`, `primary`, `secondary`, `tertiary`, `unclassified`, `residential`, `living_street`, as well as linked variants.

Pedestrian infrastructure classification uses the tags: `highway = footway | pedestrian | steps | residential` and `footway = sidewalk | crossing | path | platform | corridor | alley | track`, and `sidewalk = both | left | right`.

The calculation of cycling infrastructure ratio is analogous to the pedestrian ratio, and also extracted from OpenStreetMap . Cycling infrastructure is further classified according to its type and level of segregation, according to the following description and OSM tags:

- Cycle track or lane (*Ciclovia segregada*): Segregated or light-separated tracks exclusive for cycling.
  - `highway = cycleway`
  - `highway = path` & `bicycle = designated`
  - `segregated = yes | no`
  - `cycleway` tags containing: `separate`, `buffered_lane`, `segregated`, `lane`, or `track`.
- Advisory lane (*Via partilhada com veículos motorizados / zonas 30*): Marked cycle lanes shared with motor vehicles, such as "sharrows" or advisory lanes.
  - `cycleway` tags containing: `shared_lane`.
- Protected Active (*Via partilhada com peões*): Infrastructure shared with pedestrians but protected from motor vehicles.
  - `highway = pedestrian` & `bicycle = designated`.
  - Refinement of `highway = path | footway | pedestrian` when `foot` access is explicitly allowed on cycling paths without physical segregation.

The primary indicators are thus:

- **Cycling infrastructure ratio:**

$$R_{cyc} = \frac{total\ cycleway\ length}{road\ length}$$

- **Cycling quality ratio:**

$$R_{ped} = \frac{segregated\ cycleway\ length}{total\ cycleway\ length}$$

Where "segregated cycleway length" corresponds to the "cycle track or lane" category.

3.2.2.6. Shared mobility availability.

The number of shared mobility points (physical and virtual docks from all operators in the LMA (TML and Way2Go, 2025)) was counted and aggregated per parish and per grid cell.

3.2.3. Affordability

Affordability evaluates the capacity of households to support the monetary costs of transportation to access essential activities. IMPT uses a weighted cost by mode, in which the monetary cost of automobile/public transport travel is multiplied by each parish's modal share, and expressed as a proportion of median household income. The travel costs are estimated for one-way trips from home to the workplace during the morning peak.

3.2.3.1. Cost of driving.

3.2.3.2. Cost of public transportation.

3.2.3.3. Composite measures.

3.2.4. Safety

This dimension encompasses the disproportional exposure to externalities of the transportation system. It is entirely based on road safety data provided by Autoridade Nacional de Segurança Rodoviária (ASNR) through Transportes Metropolitanos de Lisboa (TML) (TML and Way2Go, 2025). It is important to note that only road crashes occuring *inside localities* (dentro das localidades), a category specified by the ASNR itself, are included in the index. In particular, this excludes crashes on motorways and on the two bridges crossing the Tagus river, while maintaining IMPT's spatial scope.

The indicators calculated for each geographical unit and transport mode are systematized below:

- **Total accidents (5 years):**

$$\sum(Acc_{5y})$$

- **Fatalities (30-day):**

$$\sum(VM_{30d})$$

- **Serious injuries (30-day):**

$$\sum(FG_{30d})$$

- **Slight injuries (30-day):**

$$\sum(FL_{30d})$$

- **Accident rate:**

$$\frac{\sum(Acc_{5y})}{Population} \times 1000$$

- **Fatality rate:**

$$\frac{\sum(VM_{30d})}{Population} \times 1000$$

- **Severity index (total):**

$$\frac{\sum(VM_{30d})}{\sum(Acc_{5y})}$$

- **Severity index (motorized):**

$$\frac{\sum(VM_{30d})}{Motorized\ vehicles\ involved}$$

- **Severity index (cycling):**

$$\frac{\sum(VM_{30d})}{Cyclists\ involved}$$

- **Severity index (pedestrian):**

$$\frac{\sum(VM_{30d})}{Pedestrians\ involved}$$

3.3. Normalization and Aggregation

All indicators were normalized to the range [1, 100] using min-max scaling:

For benefit indicators (higher raw value → better):

$$X_{norm} = 1 + \frac{X - X_{min}}{X_{max} - X_{min}} \times 99$$

For cost indicators (lower raw value → better):

$$X_{norm} = 1 + \frac{X_{max} - X}{X_{max} - X_{min}} \times 99$$

- **Accessibility** metrics (`access_*`) are *benefit* indicators.
- **Mobility** cost metrics (`mobility_cost_*`), public transport waiting times, all **affordability** indicators, and all **safety** indicators are *cost* indicators.

Missing values in accessibility indicators (arising from unfeasible public transportation routes) are imputed with the mean of the respective variable prior to PCA.

3.4. PCA-Based Dimension Scores

For each dimension (Accessibility, Mobility, Safety) and at the global (all-mode) and per-mode levels, a PCA is run using `FactoMineR::PCA()` at parish level.

The **first principal component (PC1)** score is used as the dimension index, after rescaling to [1, 100]. PC1 captures the dominant axis of variation among the normalised indicators.

For **Affordability**, PCA is not applied (too few indicators); instead, the normalised composite `h_transp_inc_comp` (housing + transport cost as share of income, modal-share weighted) is used directly as the Affordability Index.

For **Safety per mode**, an equal-weight average of the severity index (per mode) and total vehicles involved (per mode) is used instead of PCA.

For **Mobility - Car** PCA is not applied (too few indicators); instead, we assumed an equal-weight average of the normalized composite `mobility_commuting_avg_tt_car` and `road_length`.

3.5. Composite IMPT Scores

Three aggregation methods are computed for the global IMPT and each per-mode IMPT:

1. **Geometric mean** (equal weights, Figure 1a):

$$IMPT_{geom} = \left(\prod_d (S_d + 1) \right)^{1/n} - 1$$

Partially compensatory: penalizes dimensions with very low values non-linearly. Highlights the most critical situations, where at least one dimension is severely deficient.

2. **Arithmetic mean** (equal weights, Figure 1b):

$$IMPT_{avg} = \frac{1}{n} \sum_{d} S_d$$

Compensatory: high vulnerability in one dimension can be offset by good performance in others. Suitable for setting aggregate policy targets and communicating to non-specialist audiences.

3. **Entropy-weighted geometric mean** (Figure 1c):

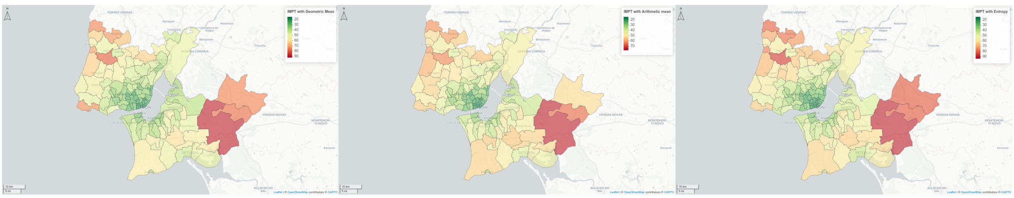
$$IMPT_{entropy} = \prod_{d} (S_d + 1)^{w_d} - 1$$

where weights w_d are derived from the **entropy weighting method**:

Weights are determined by the empirical dispersion of the dimensions across parishes: the dimension that varies most among territories receives greater weight. Maximises territorial contrast and is useful for competitive prioritisation of interventions.

$$w_d = \frac{d_d}{\sum_j d_j}, \quad d_d = 1 - e_d, \quad e_d = -k \sum_{i} p_{id} \ln(p_{id})$$

with $k = 1/\ln(n)$ and p_{id} being the proportion of the dimension score for area i.



(a) Geometric mean (b) Arithmetic mean (c) Entropy-weighted mean

Figure 1: IMPT results for different aggregation methods for all modes (with Navegante)

4. Results

Figure 2 is generated using an R chunk.

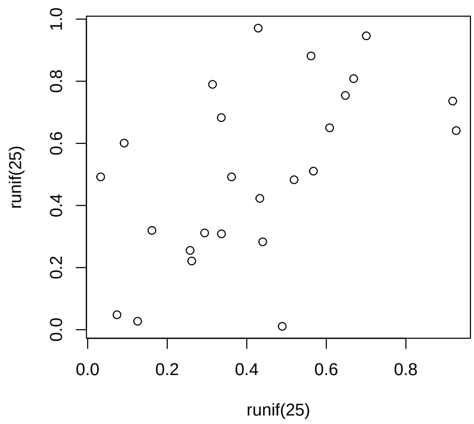


Figure 2: A meaningless scatterplot

5. Discussion

5.1. Limitations

Tables can also be generated using R chunks, as shown in Table 4 example.

```
knitr::kable(head(mtcars)[,1:4])
```

```
Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
Use 'xfun::attr2()' instead.
See help("Deprecated")
```

```
Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
Use 'xfun::attr2()' instead.
See help("Deprecated")
```

Table 4: Caption centered above table

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93

Table 4: Caption centered above table

	mpg	cyl	disp	hp
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
Valiant	18.1	6	225	105

6. Conclusions

This is the best paper ever.

References

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